Program 5
Introduction to Jenkins: What is Jenkins? Installing Jenkins on Local or Cloud Environment, Configuring Jenkins for First Use

Aim

To understand Jenkins and perform basic installation and configuration of Jenkins.

What is Jenkins?

Jenkins is an **open-source automation server** used in DevOps to automate software development tasks such as:

- Building applications
- Running tests
- Deploying applications

It is mainly used to implement **Continuous Integration (CI)** and **Continuous Deployment (CD)**.

In simple terms, Jenkins automatically builds and tests code whenever developers make changes.

Why Jenkins is Used

• Automates software build process
• Detects errors early in development
• Saves developer time
• Supports many plugins for integration with tools like Git, Docker, Kubernetes etc.

Jenkins Installation Steps

Step 1 Install Java

Jenkins requires Java.

Check Java installation:

```
java -version
```

Install **OpenJDK 17 or later** if Java is not available.

Step 2 Download Jenkins

Download Jenkins from the official website:

https://www.jenkins.io/download/

Download **Jenkins LTS (.msi)** for Windows.

Step 3 Install Jenkins

Run the installer and complete installation.

Jenkins runs on default port:

```
http://localhost:8080
```

Open the above URL in the browser.

Initial Jenkins Configuration

Step 1 Unlock Jenkins

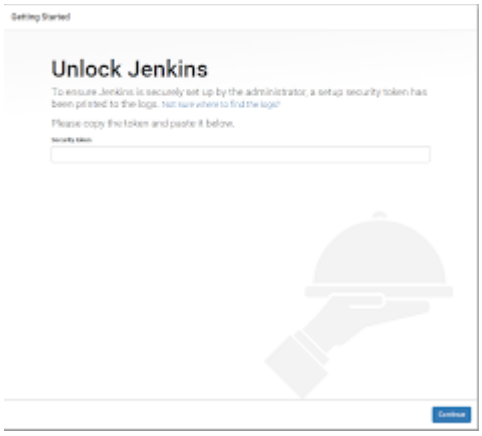
When Jenkins starts for the first time, it asks for an **Admin Password**.

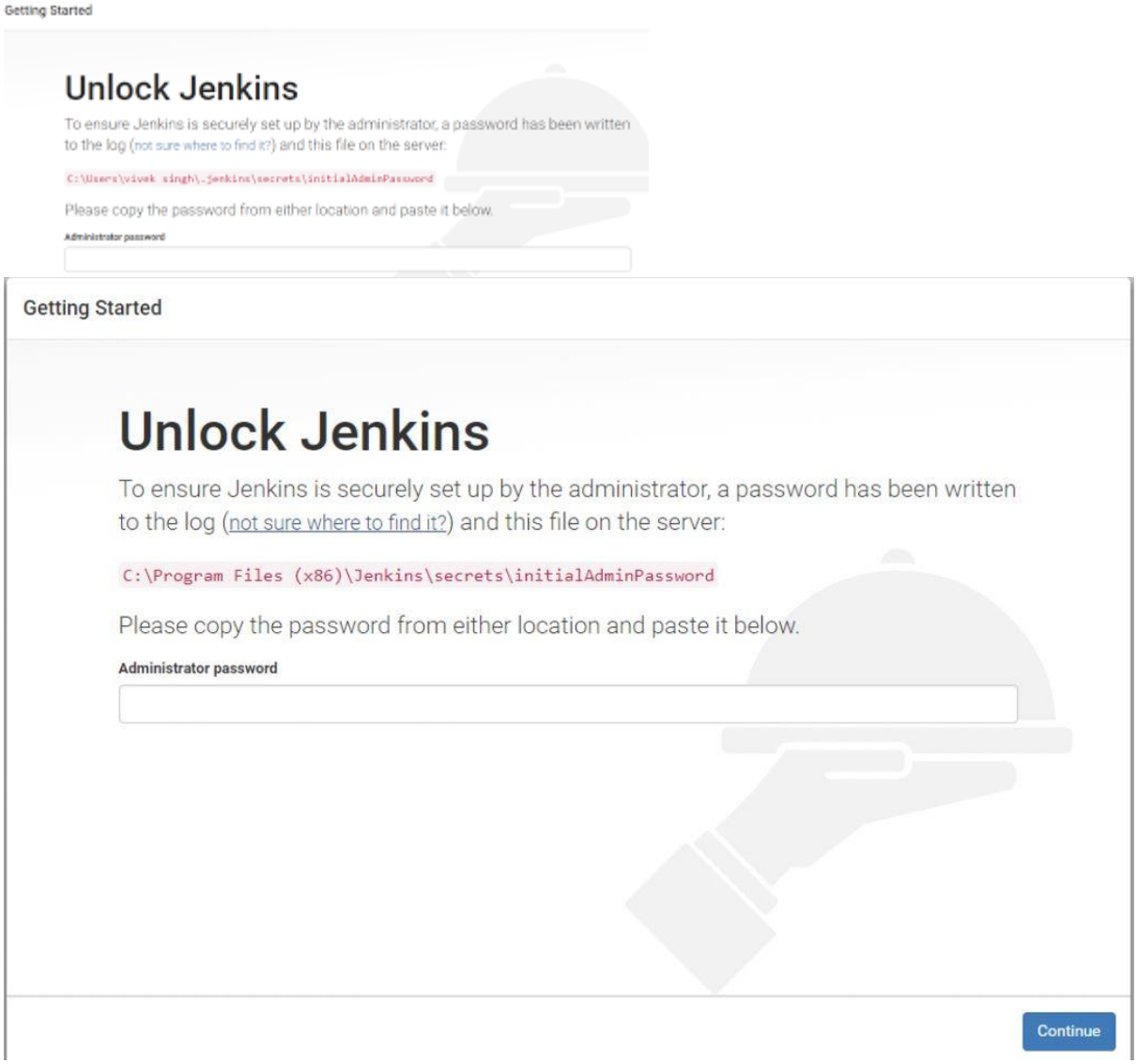
Password file location:

```
C:\Program Files\Jenkins\secrets\initialAdminPassword
```

Copy the password and paste it in the browser.

Students should capture this screen.





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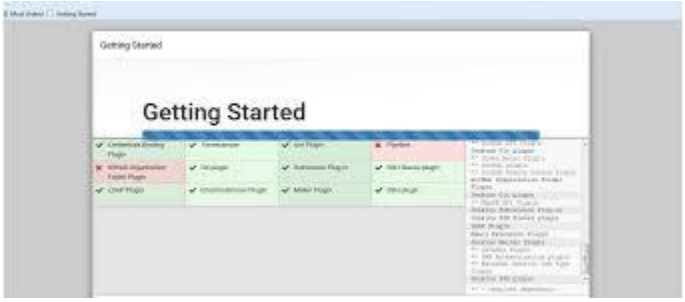
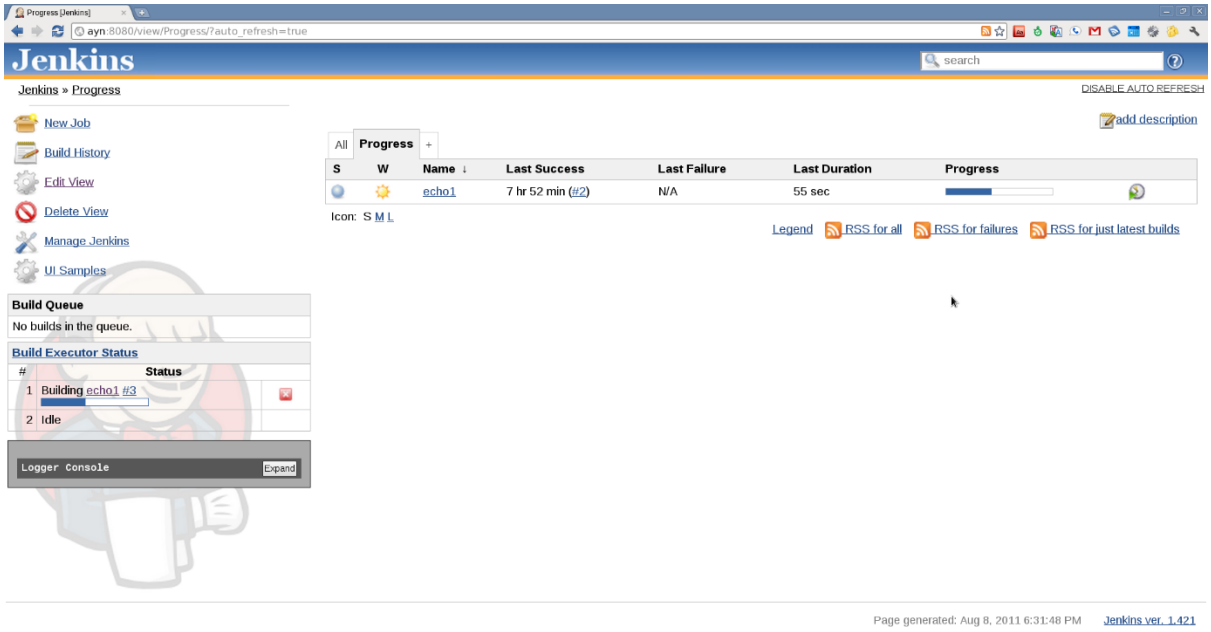
Step 2 Install Plugins

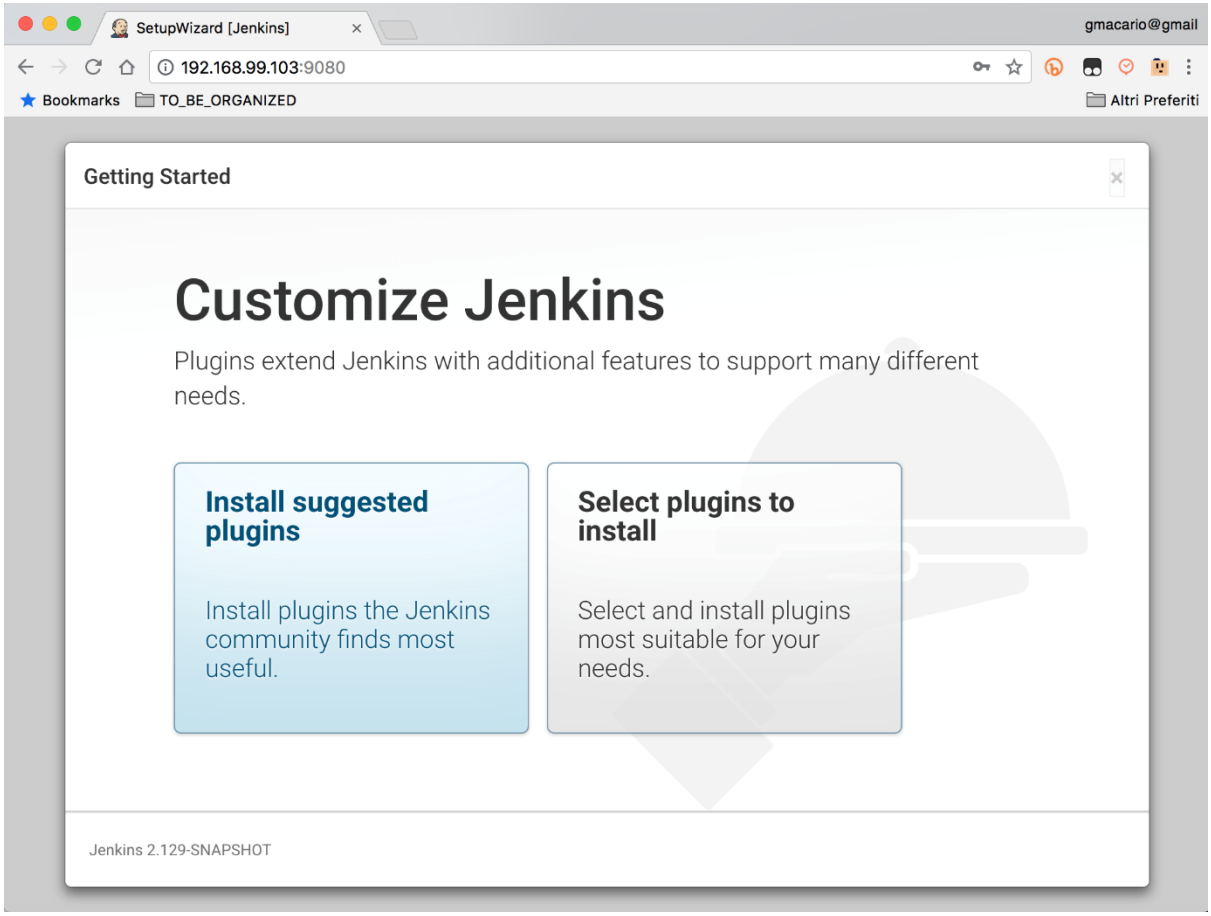
Select:

Install Suggested Plugins

This installs commonly used Jenkins plugins.




Jenkins
Jenkins » Progress
DISABLE AUTO REFRESH
New Job
Build History
Edit View
Delete View
Manage Jenkins
UI Samples
add description
All
Progress
S
W
Name
Last Success
Last Failure
Last Duration
Progress
echo1
7 hr 52 min (#2)
N/A
55 sec
Icon: S M L
Legend
RSS for all
RSS for failures
RSS for just latest builds
Build Queue
No builds in the queue.
Build Executor Status
Status
1 Building echo1 #3
2 Idle
Logger Console
Expand
Page generated: Aug 8, 2011 6:31:48 PM
Jenkins ver. 1.421

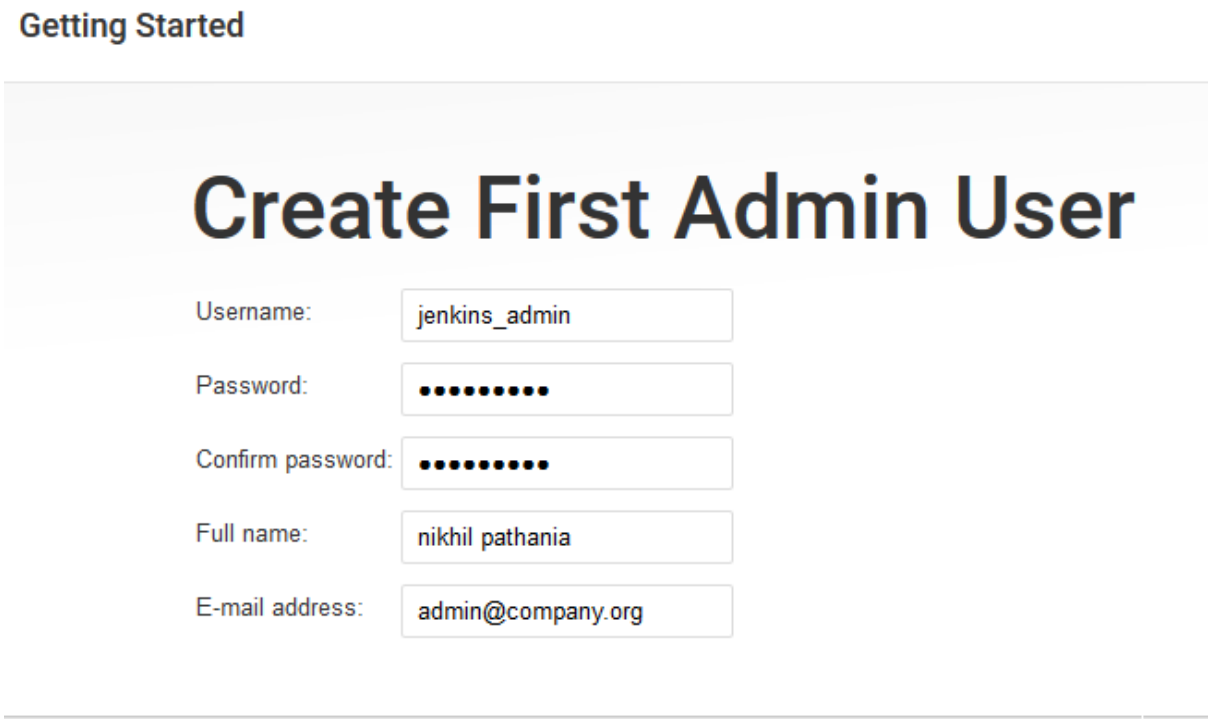

SetupWizard [Jenkins]
gmacario@gmail
192.168.99.103:9080
Bookmarks
TO_BE_ORGANIZED
Altri Preferiti
Getting Started
Customize Jenkins
Plugins extend Jenkins with additional features to support many different needs.
Install suggested plugins
Install plugins the Jenkins community finds most useful.
Select plugins to install
Select and install plugins most suitable for your needs.
Jenkins 2.129-SNAPSHOT

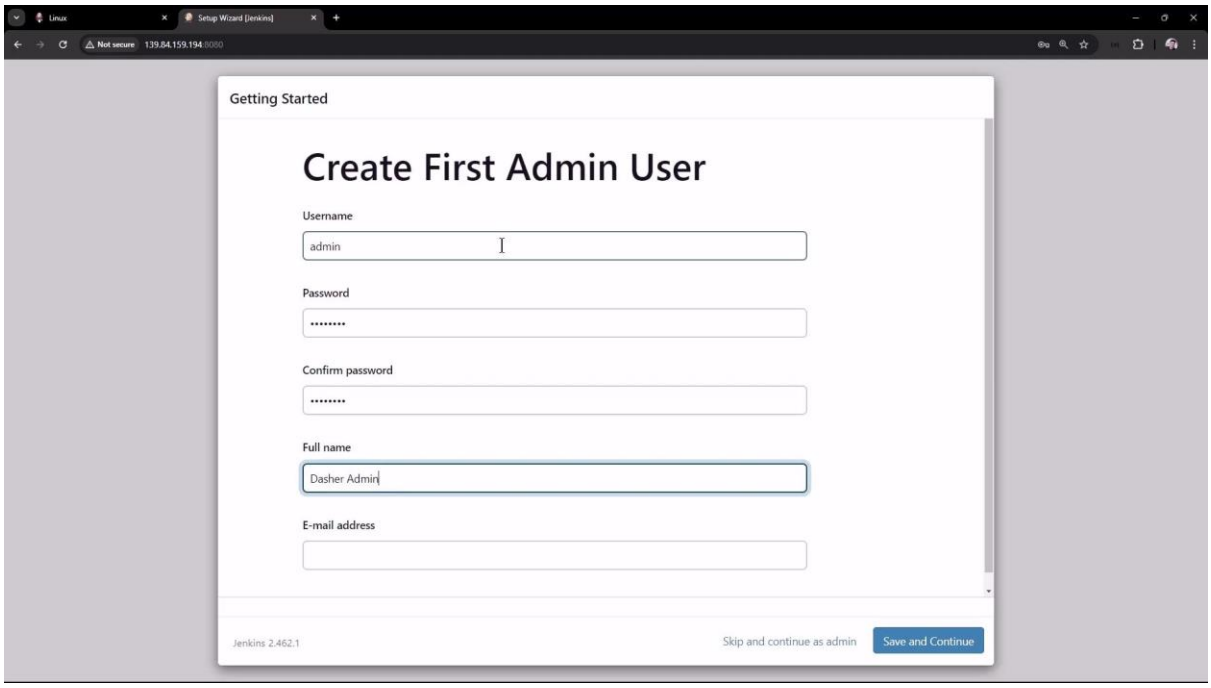
Step 3 Create Admin User

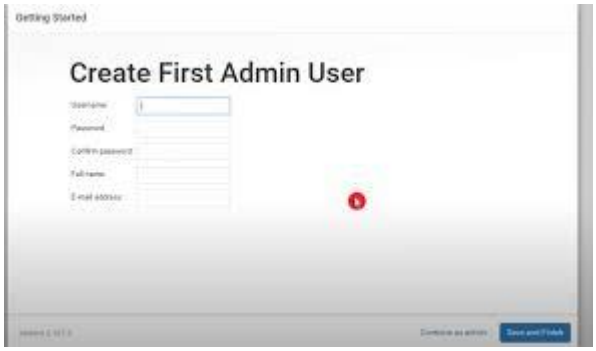
Create the first administrator account for Jenkins.

Example:

Username: admin
Password: admin123







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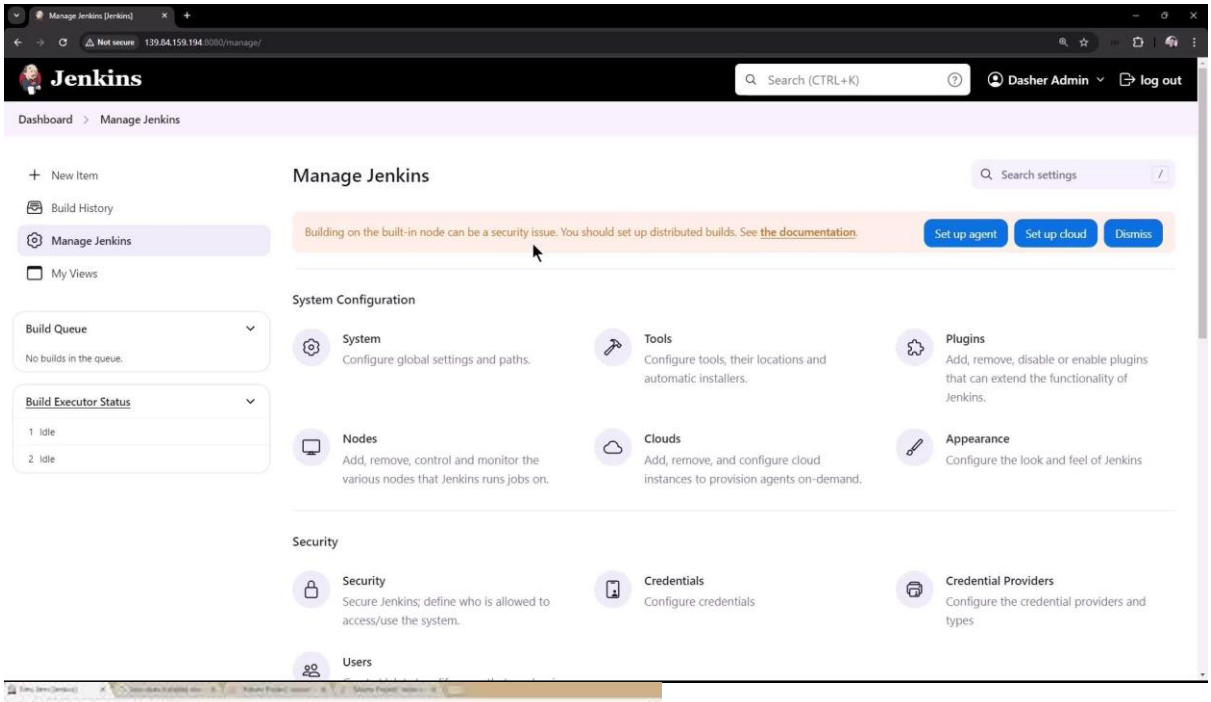
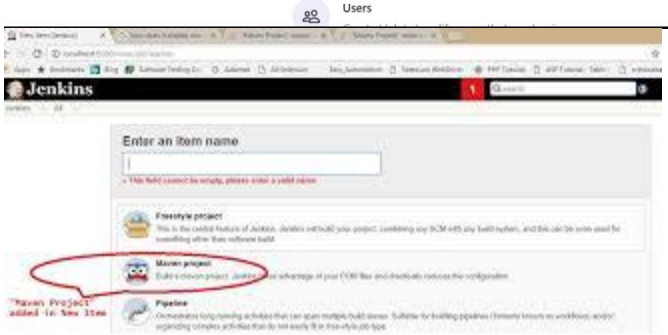
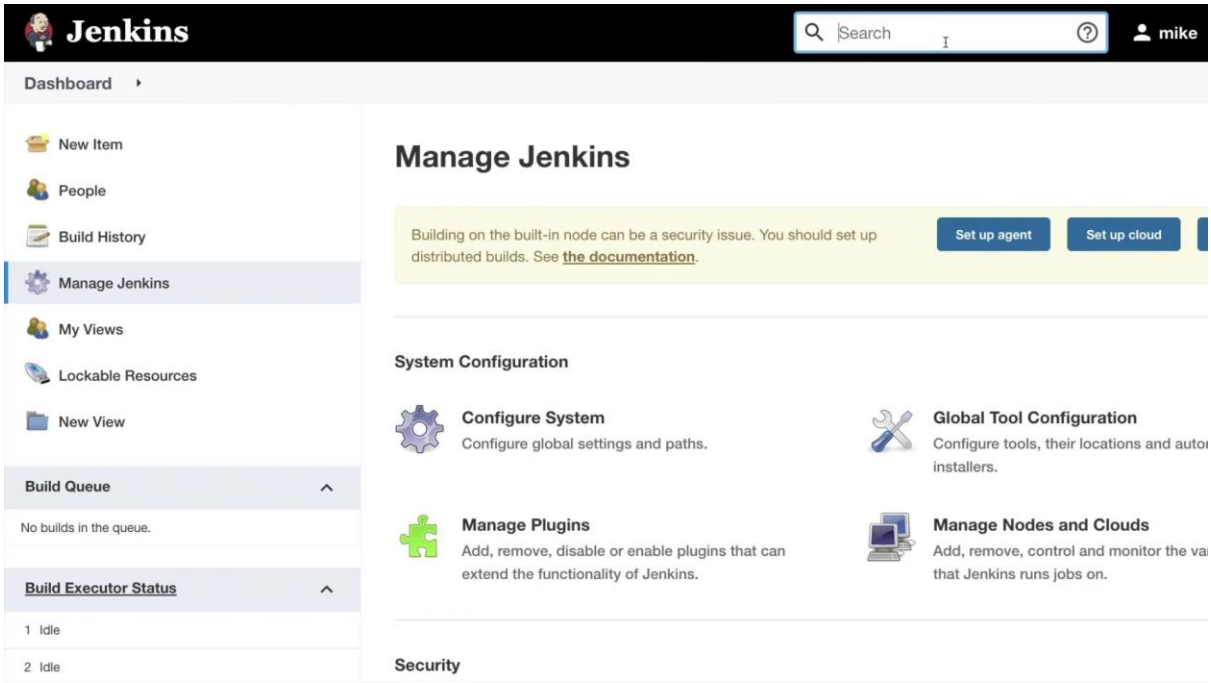
Jenkins Dashboard

After configuration, Jenkins dashboard will appear.

Students should capture the dashboard screenshot.






Jenkins
Search
mike
Dashboard
New Item
People
Build History
Manage Jenkins
My Views
Lockable Resources
New View
Build Queue
No builds in the queue.
Build Executor Status
1 Idle
2 Idle
Manage Jenkins
Building on the built-in node can be a security issue. You should set up distributed builds. See the documentation.
Set up agent
Set up cloud
System Configuration
Configure System
Configure global settings and paths.
Global Tool Configuration
Configure tools, their locations and auto
installers.
Manage Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
Manage Nodes and Clouds
Add, remove, control and monitor the va
that Jenkins runs jobs on.
Security